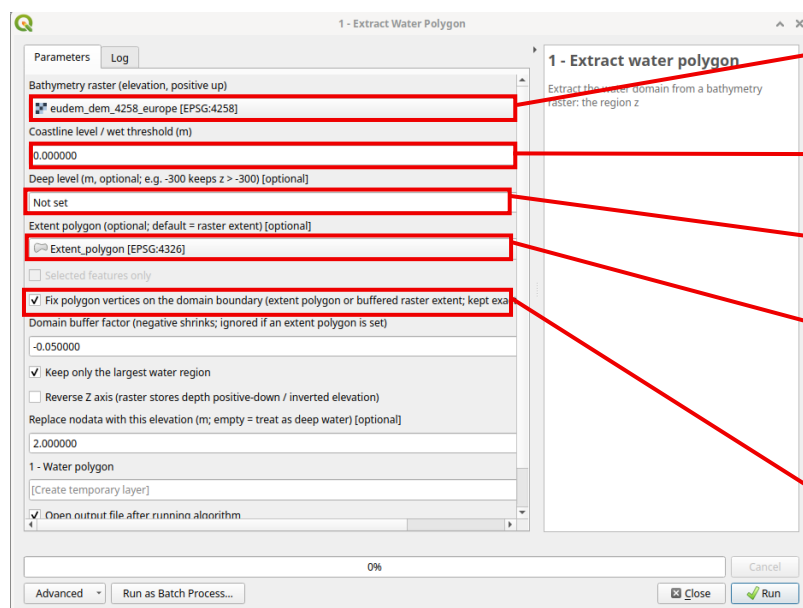
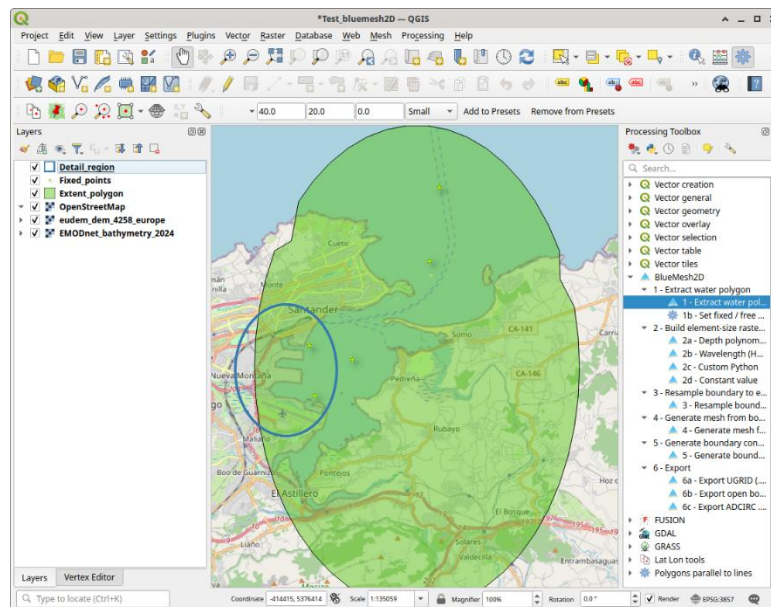


Tutorial BlueMesh2D QGIS

1 – Extract water polygon

By default, you need to define a polygon representing the study domain. If you do not provide one, a buffer inside the input raster will be used instead.

In the example dataset, the domain is the Bay of Santander, on the northern coast of Spain: the Extent_polygon (green) layer encloses the bay and a stretch of adjacent open sea, so the offshore boundary sits well away from the area of interest. An optional Detail_region (blue) polygon is placed over the Port of Santander, in the inner part of the bay, where a finer resolution is wanted.



Raster used to extract the coastline (typically a DEM).

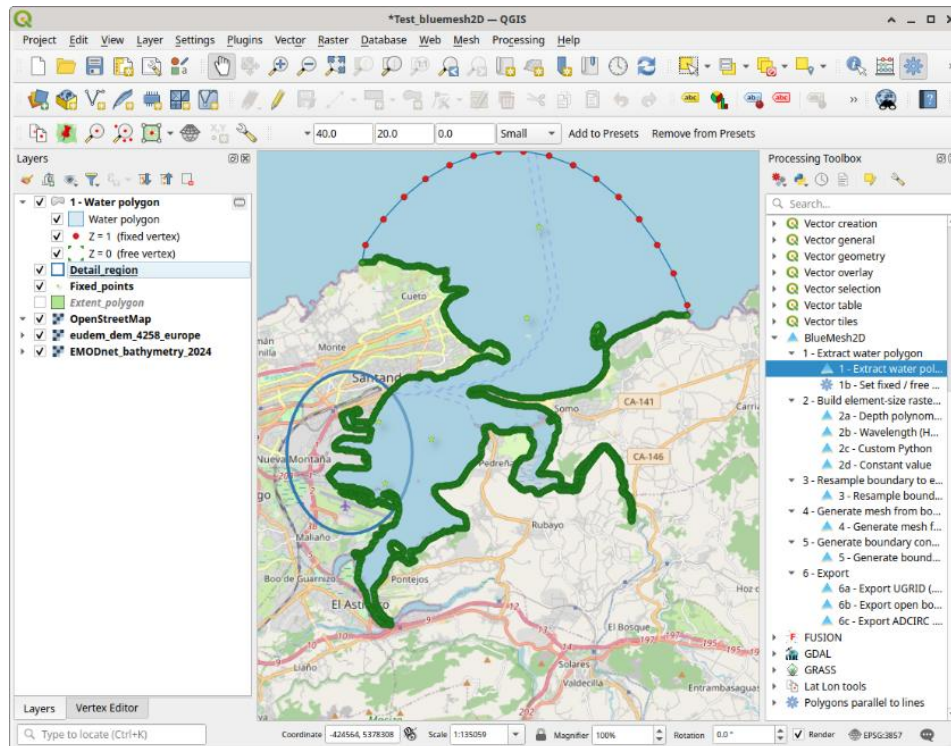
Elevation used for coastline extraction (typically Mean Sea Level, MSL).

Second depth contour.

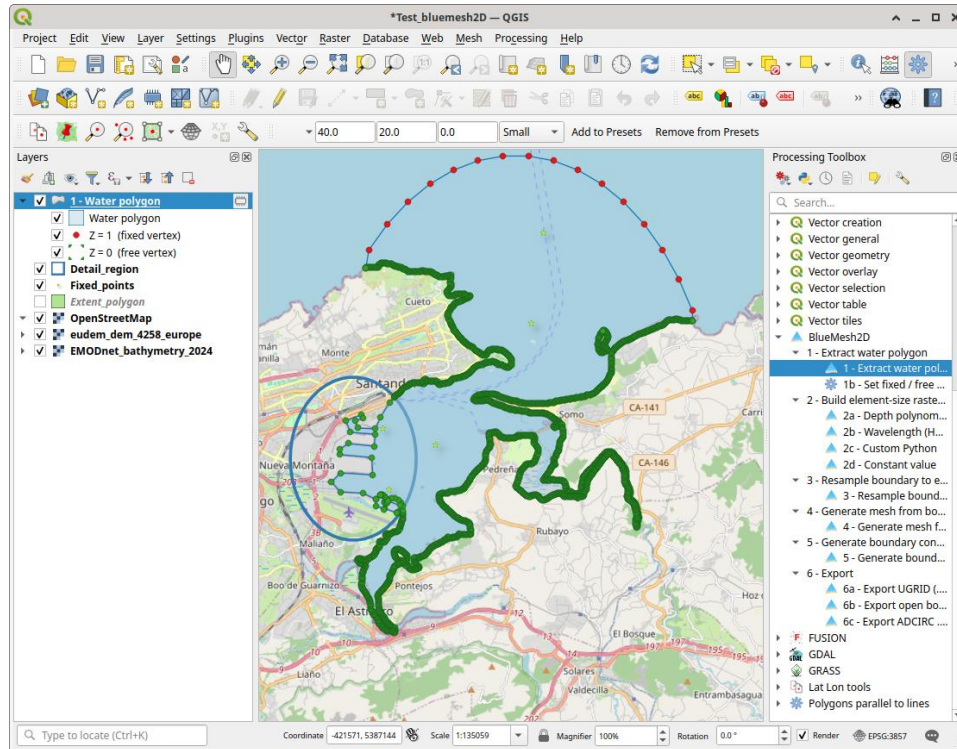
Extent polygon used to generate the selected depth contour.

By default, the vertices of the extent polygon are fixed to preserve features such as the corners of a rectangular domain.

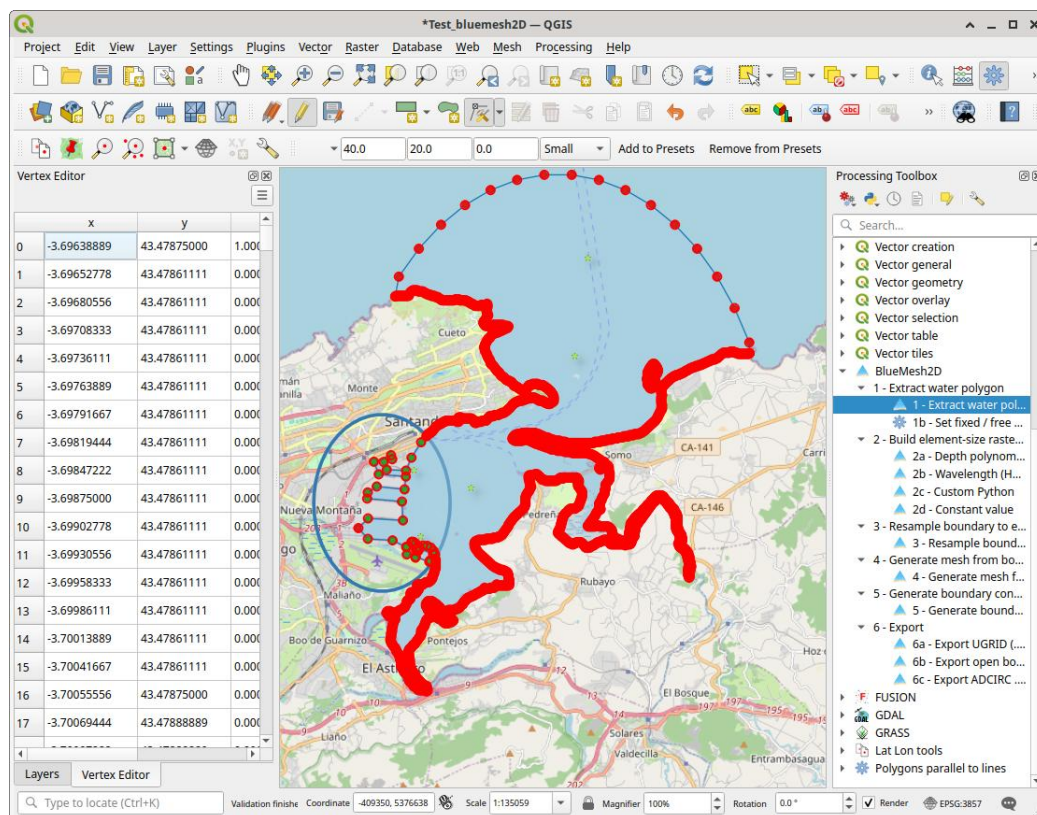
A polygon representing the bathymetric cut can then be generated. Either a single depth contour, corresponding to the coastline, or two depth contours may be defined. When two contours are specified, the polygon is generated between the coastline and the selected depth. Alternatively, user-provided data may be used.



The extracted contour is displayed with fixed vertices shown in red and non-fixed vertices shown in green. Fixed vertices are preserved in the final contour after resampling. Non-fixed vertices may be repositioned during the resampling process, although the contour segments between them are retained.

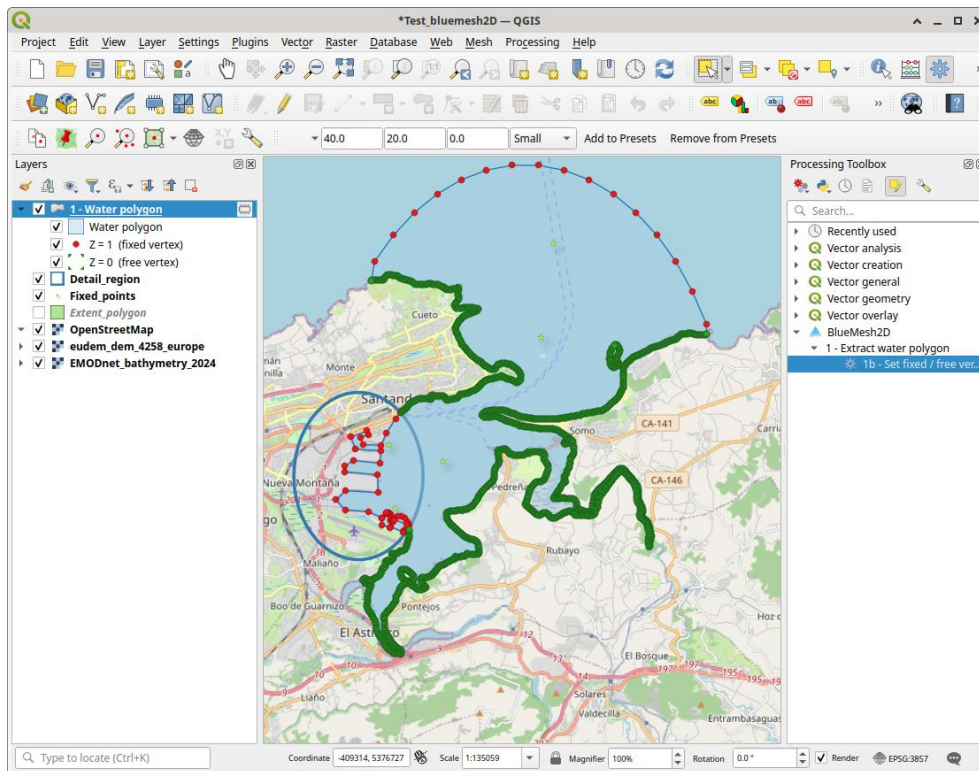
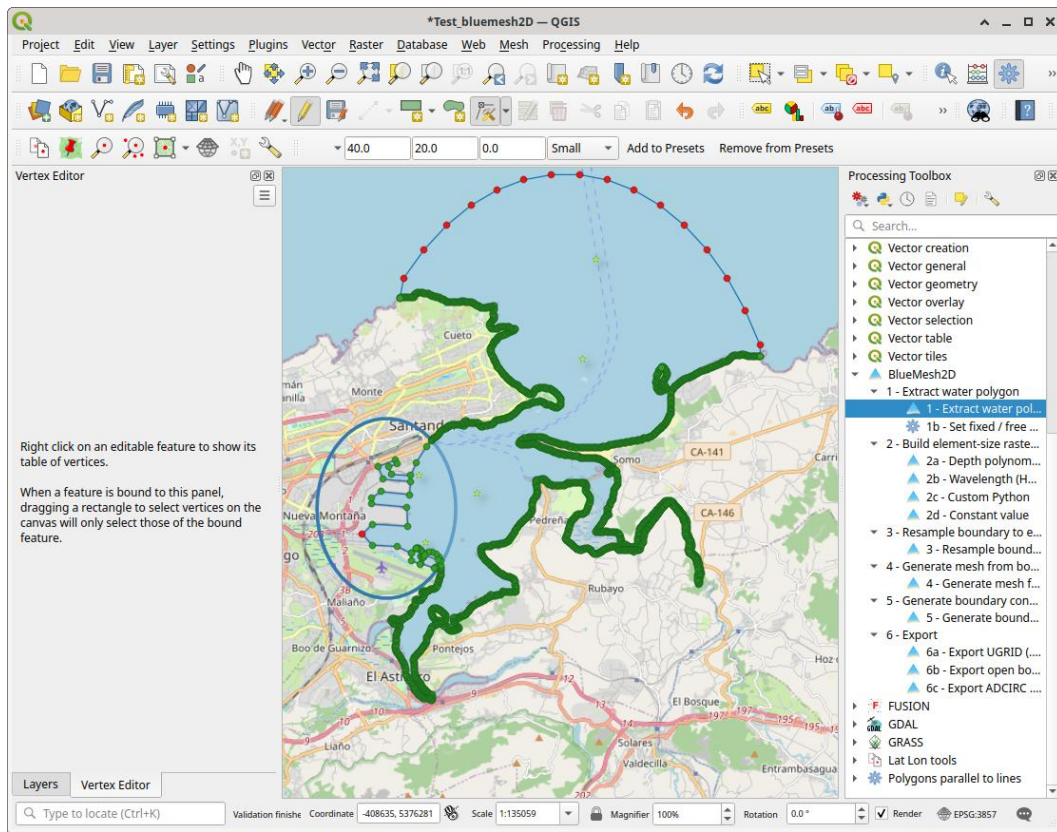


The contour can be edited as a standard polygon using the Vertex Editor. In areas of interest, the polygon may be modified or simplified to provide a better representation of the actual topography.

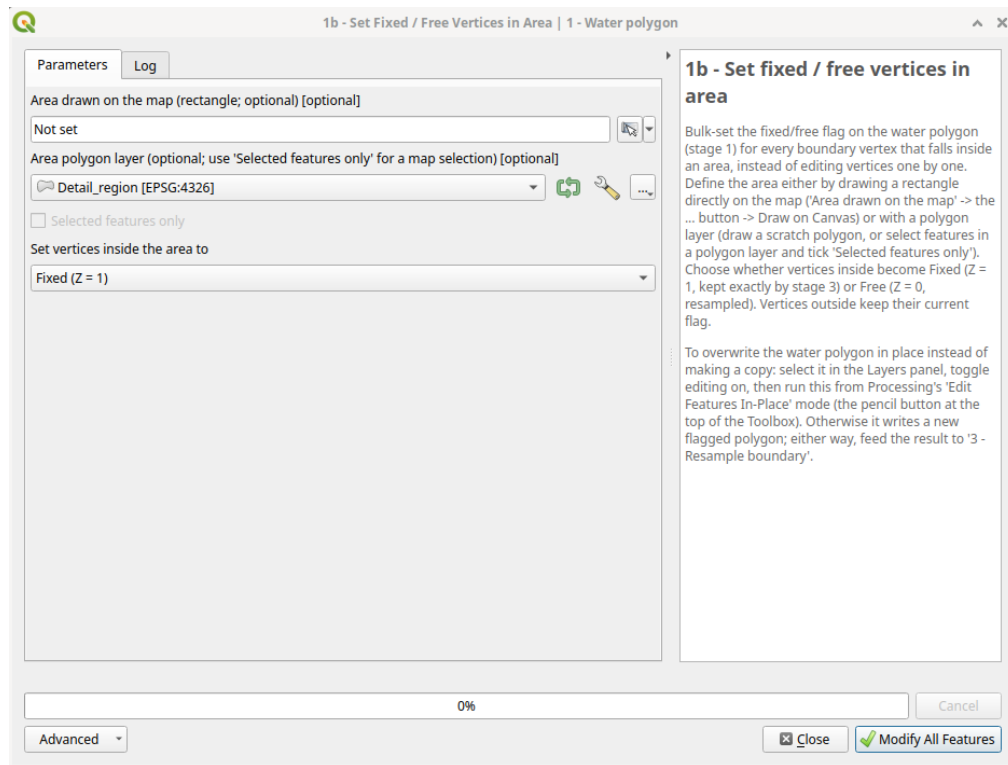


The properties of each vertex can also be edited using the Vertex Editor. The fixed status of a vertex is stored in its Z value and can therefore be viewed or modified directly within the

Vertex Editor. With the polygon in edit mode (pencil tool), right-click on the map and select the vertex to be edited. The vertex's Z value can then be modified in the Vertex Editor.



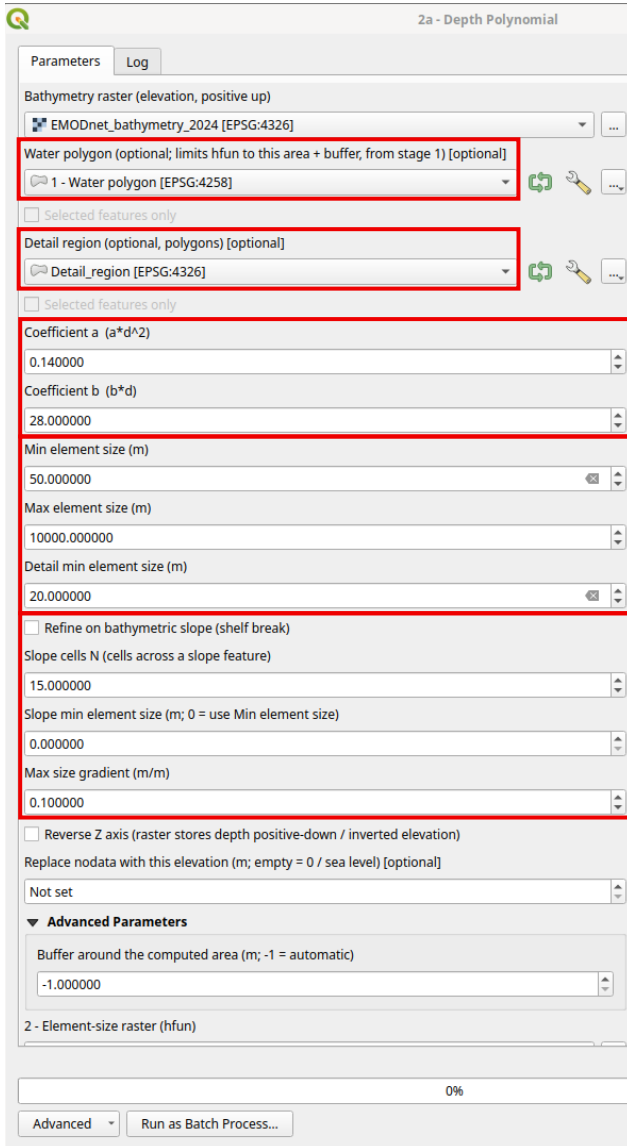
Multiple vertices can also be edited simultaneously. With the layer in edit mode, click Edit Features In-Place and then select Set Fixed/Free Vertices.



The Set Fixed/Free Vertices tool allows the status of multiple vertices to be changed simultaneously. Vertices can be selected within either a user-defined area drawn on the map or a predefined polygon from a polygon layer.

2 – Build element-size raster (hfun)

The mesh is generated according to the target cell sizes stored in the HFUN layer. Several mesh-size functions are available, including a polynomial law (recommended for energetic models) and a wavelength-based function (recommended for phase-averaged wave models models).



Full domain extent

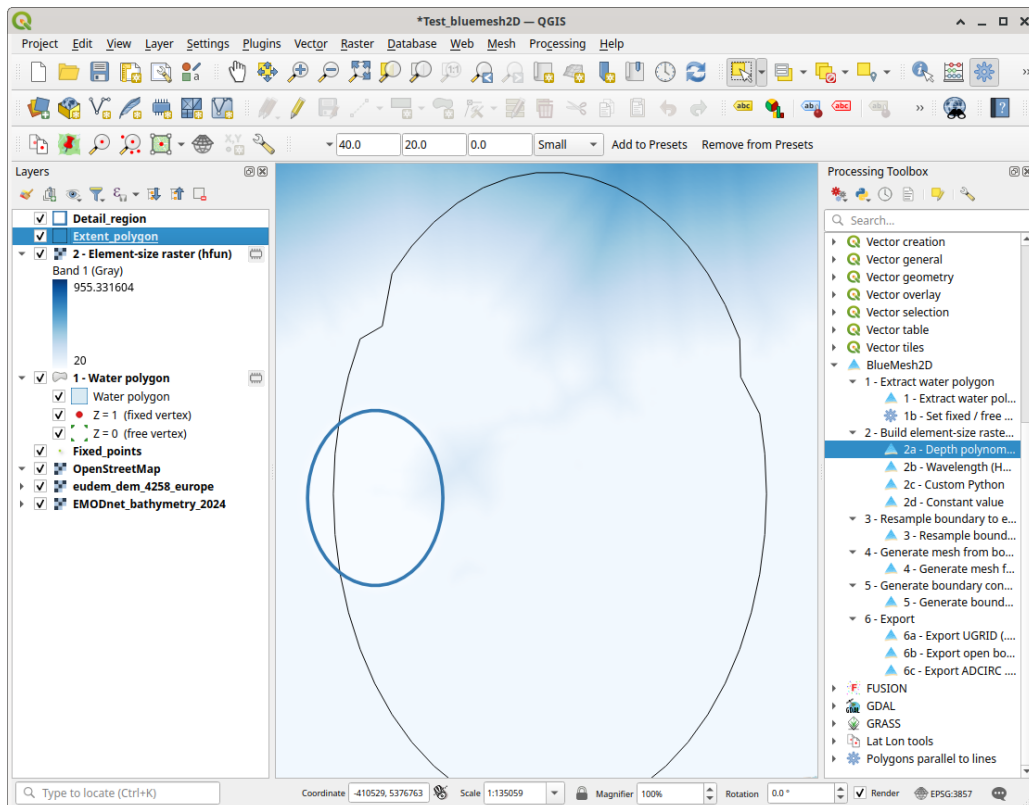
Detail region (area where detail min element size will be used)

Parameters of the polynomial functions

Clipping thresholds

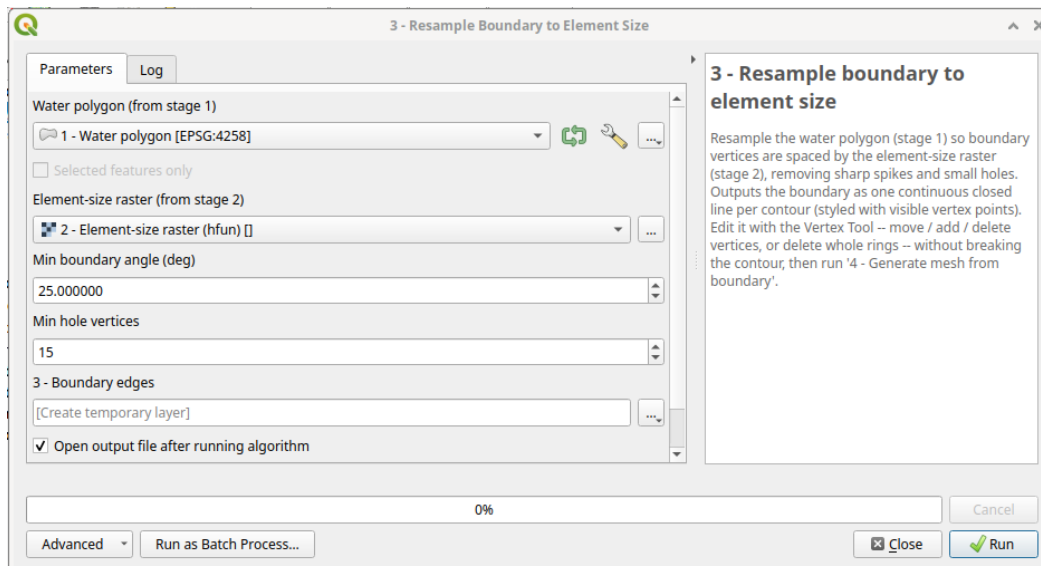
Depth gradient refinement

Applying a mesh-size function generates a raster of target cell sizes. The functions currently implemented depend only on water depth and bathymetric gradients. However, custom mesh-size functions can also be defined through a user-defined expression.

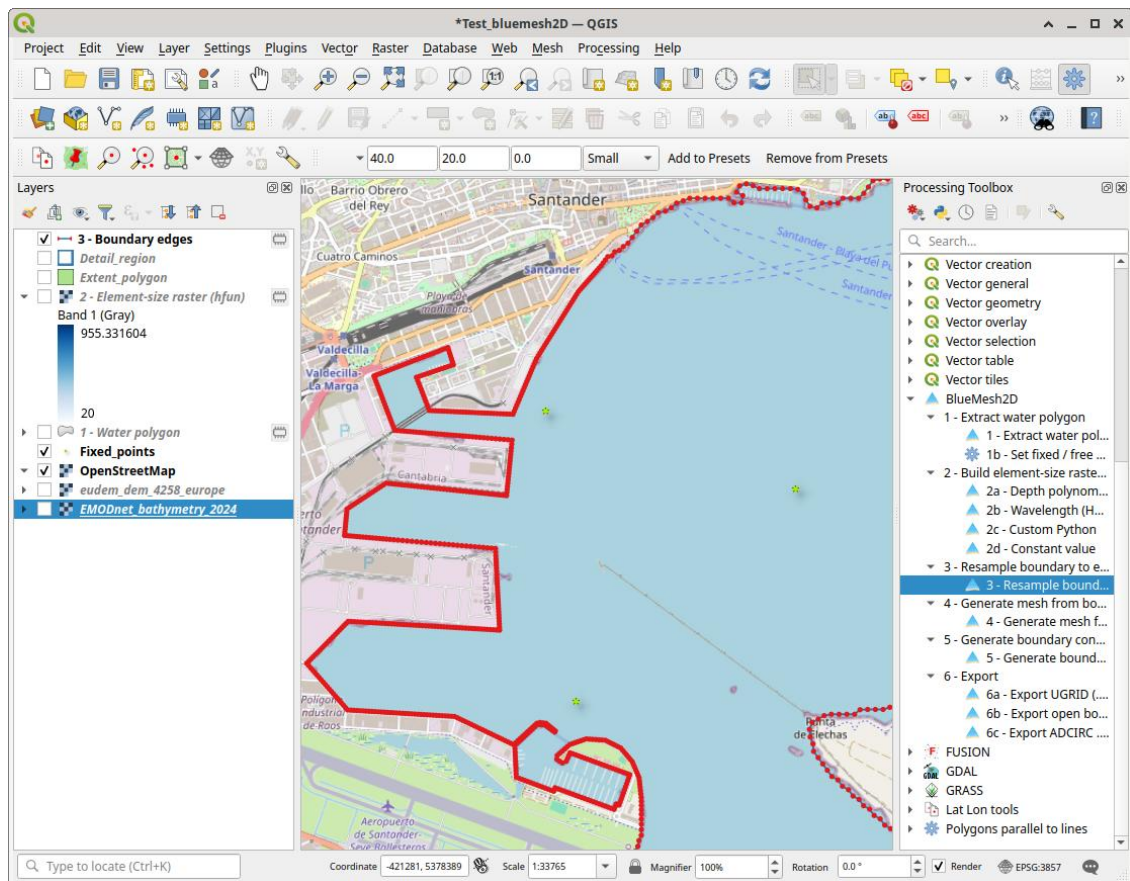
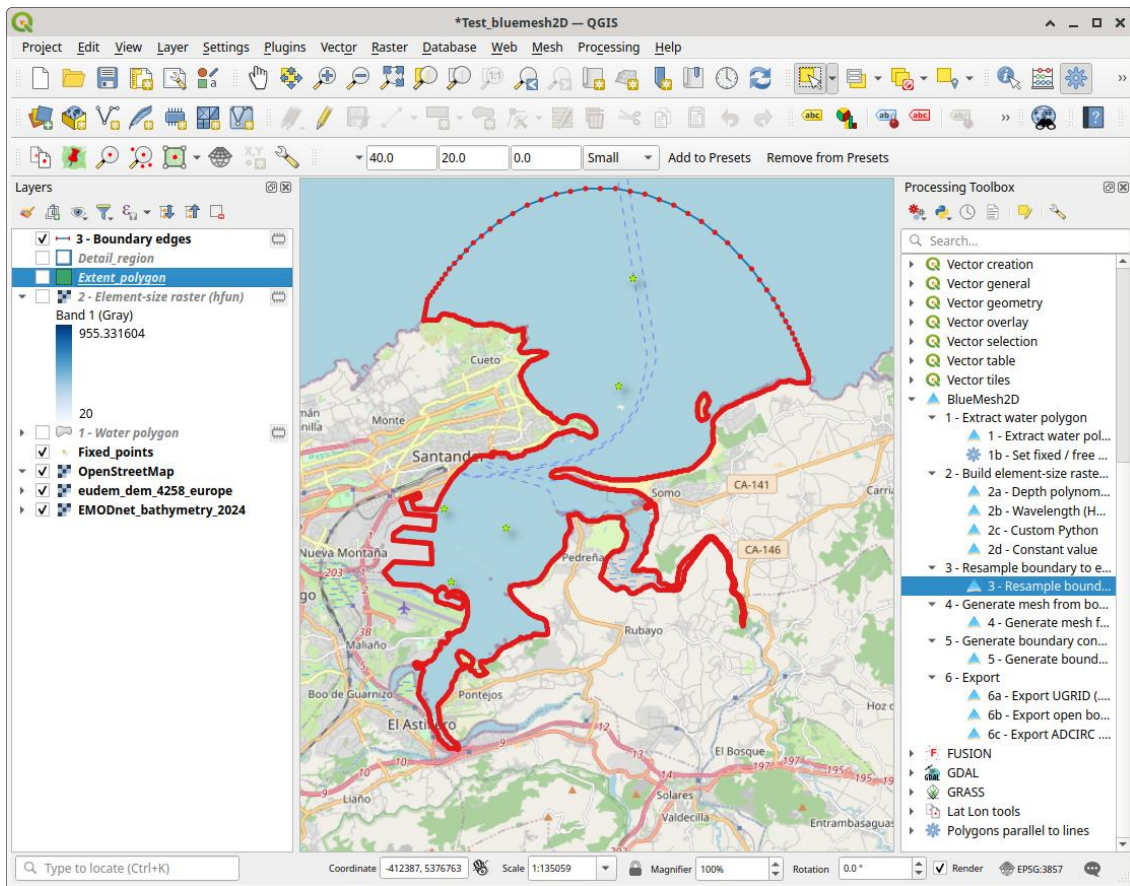


An example of the target cell-size raster is shown below. This raster is used to resample the polygon before mesh generation. The objective is to ensure that the sampling of the boundary edges is consistent with the target mesh resolution.

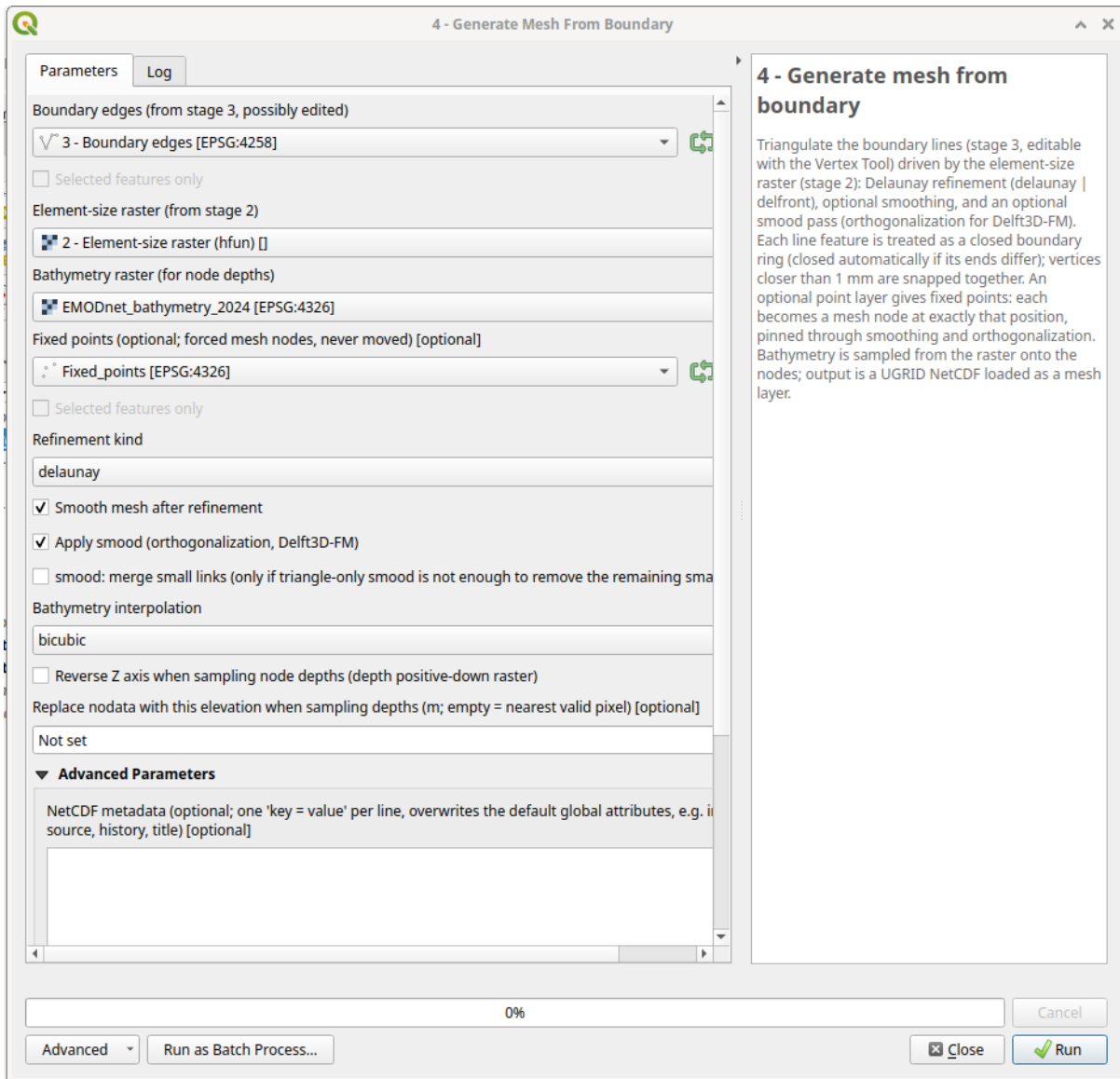
3- Resample boundary to element size



The figure below shows the resampled polygon used to generate the mesh boundaries. Several parameters are available to control the resampling process, including the minimum number of vertices required to distinguish an island from a hole generated by shallow areas, and the minimum angle allowed between three consecutive vertices. The latter prevents the creation of excessively sharp angles, which can lead to instabilities during mesh generation.



4 – Generate mesh from boundary



4 - Generate mesh from boundary

Triangulate the boundary lines (stage 3, editable with the Vertex Tool) driven by the element-size raster (stage 2): Delaunay refinement (delaunay | delfront), optional smoothing, and an optional smood pass (orthogonalization for Delft3D-FM). Each line feature is treated as a closed boundary ring (closed automatically if its ends differ); vertices closer than 1 mm are snapped together. An optional point layer gives fixed points: each becomes a mesh node at exactly that position, pinned through smoothing and orthogonalization. Bathymetry is sampled from the raster onto the nodes; output is a UGRID NetCDF loaded as a mesh layer.

Parameters | **Log**

Boundary edges (from stage 3, possibly edited)

3 - Boundary edges [EPSG:4258]

☐ Selected features only

Element-size raster (from stage 2)

2 - Element-size raster (hfun) []

Bathymetry raster (for node depths)

EMODnet_bathymetry_2024 [EPSG:4326]

Fixed points (optional; forced mesh nodes, never moved) [optional]

Fixed_points [EPSG:4326]

☐ Selected features only

Refinement kind

delaunay

☒ Smooth mesh after refinement

☒ Apply smood (orthogonalization, Delft3D-FM)

☐ smood: merge small links (only if triangle-only smood is not enough to remove the remaining small links)

Bathymetry interpolation

bicubic

☐ Reverse Z axis when sampling node depths (depth positive-down raster)

Replace nodata with this elevation when sampling depths (m; empty = nearest valid pixel) [optional]

Not set

Advanced Parameters

NetCDF metadata (optional; one 'key = value' per line, overwrites the default global attributes, e.g. in source, history, title) [optional]

0%

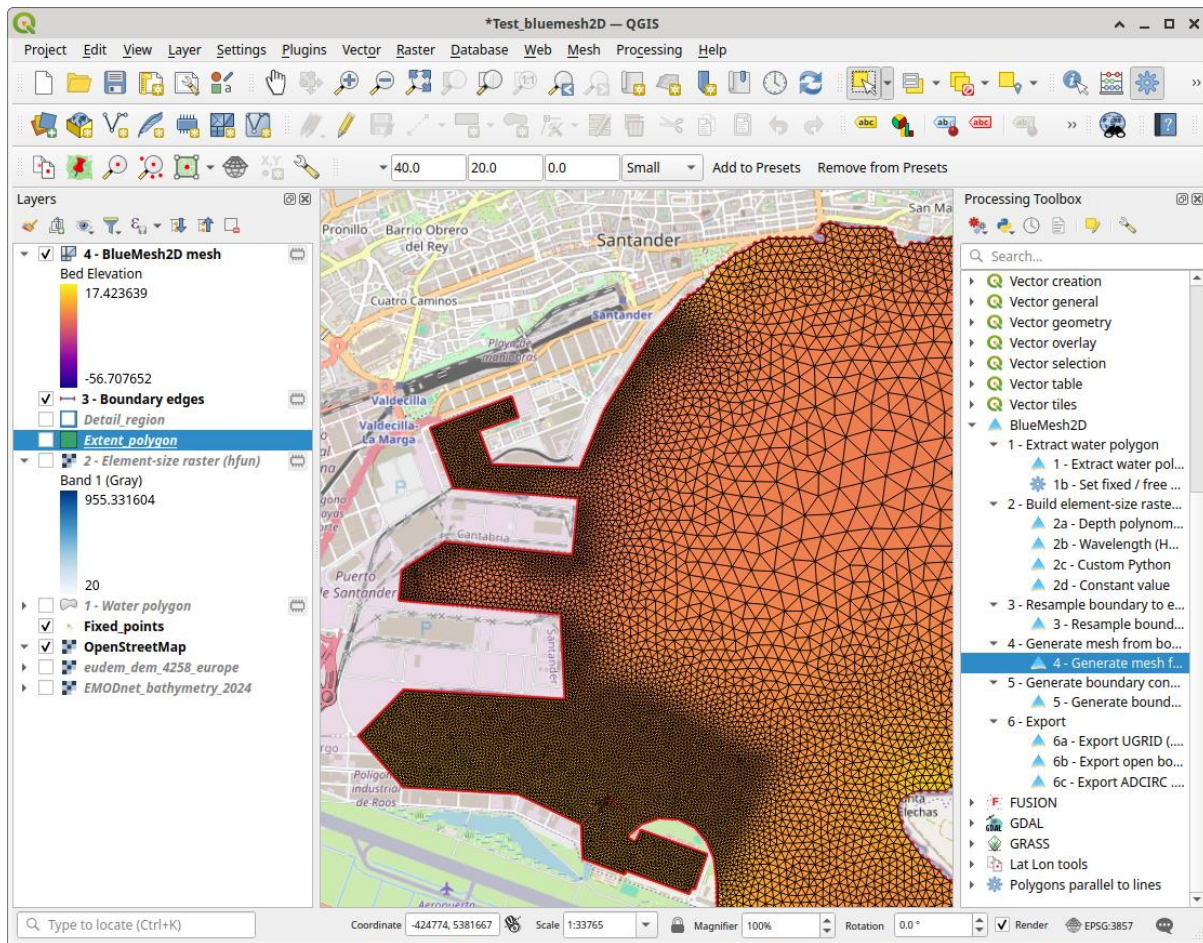
Advanced | Run as Batch Process...

Close Run

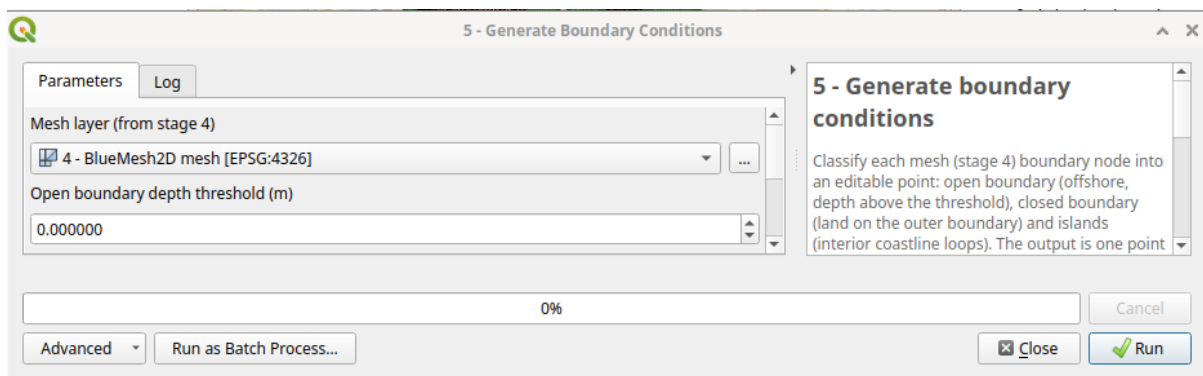
The mesh generation panel is shown below. The mesh is generated from the boundary edges and the target cell-size (HFUN) raster. Two mesh generation algorithms are available: the classical Delaunay refinement method and the Frontal-Delaunay refinement method.

Once the initial mesh has been generated, it can be improved using several optional post-processing steps. A DistMesh-style optimization can be applied to smooth the mesh, followed by a Delft3D-FM orthogonalization pass ported from the Deltares MeshKernel library. If the orthogonalization process fails to converge, the Merge Quad-Tri option can be enabled to convert problematic triangles into quadrilateral elements, improving mesh quality and convergence.

The bathymetry interpolation method used to assign depth values to the mesh nodes can also be selected.



5 – Generate boundary conditions



This algorithm assigns a boundary-condition type to every boundary node of the mesh. It produces an editable point layer with one point per boundary node, carrying the type in the btype attribute (open, closed or island) and the sampled depth in depth.

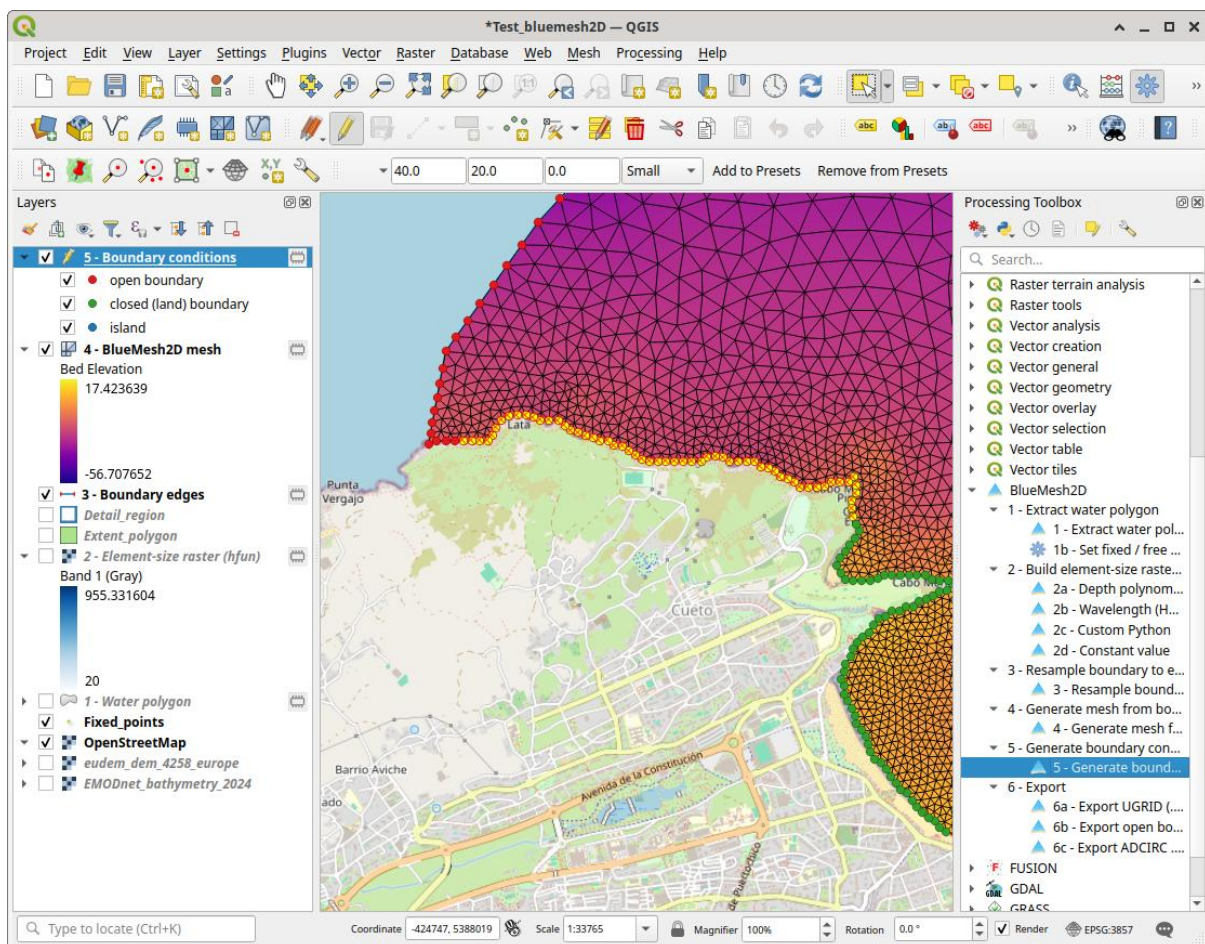
It takes a single parameter, Open boundary depth threshold (m). This is a depth, positive downwards, and it is compared node by node against the bathymetry interpolated onto the mesh at stage 4. It must be deeper than the level at which the coastline was generated in stage

1 (Coastline level / wet threshold); otherwise, every boundary node is deeper than the threshold and the entire outer boundary comes out as open. For a coastline generated at 0 m, a threshold of a few metres (e.g. 5 m) is a sensible starting point.

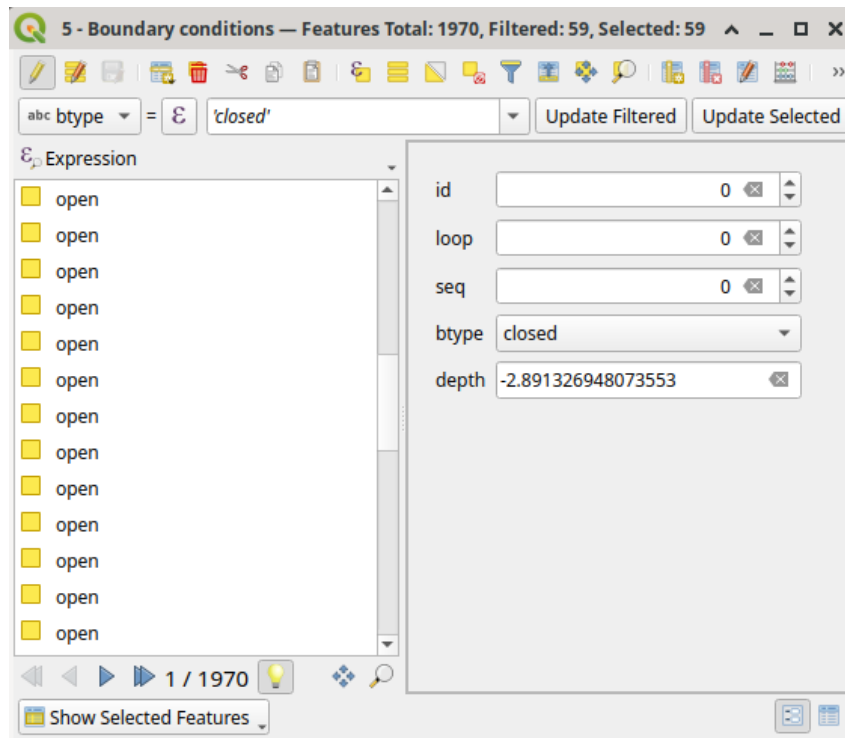
Each boundary node is classified as follows:

- island — the node lies on a boundary loop that is enclosed within another, larger loop. These interior coastlines are always islands, whatever their depth: for them the threshold is not evaluated at all.
- open — the node lies on the outer boundary, and its depth is deeper than the threshold. This is the open sea, where forcing is applied.
- closed — the node lies on the outer boundary, and its depth is shallower than or equal to the threshold. This is land.

This method is not infallible, so incorrectly classified nodes can be corrected manually.

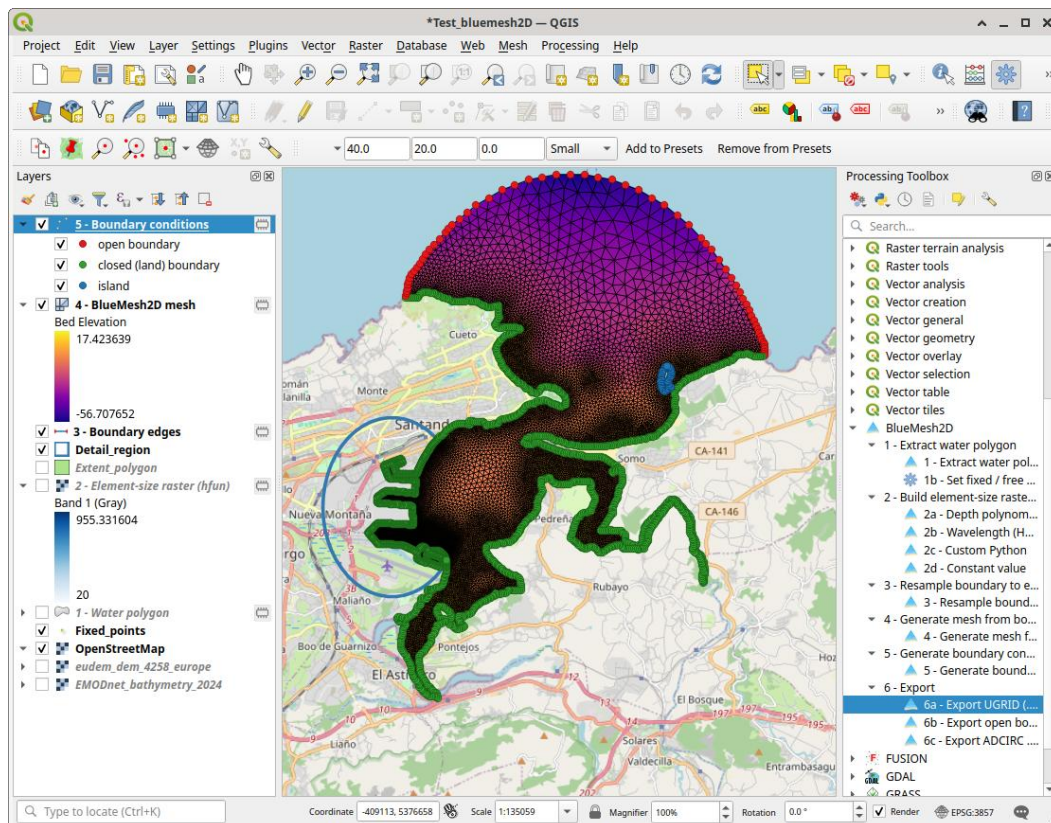


The type of each boundary node can be edited in the attribute table while the layer is in edit mode.

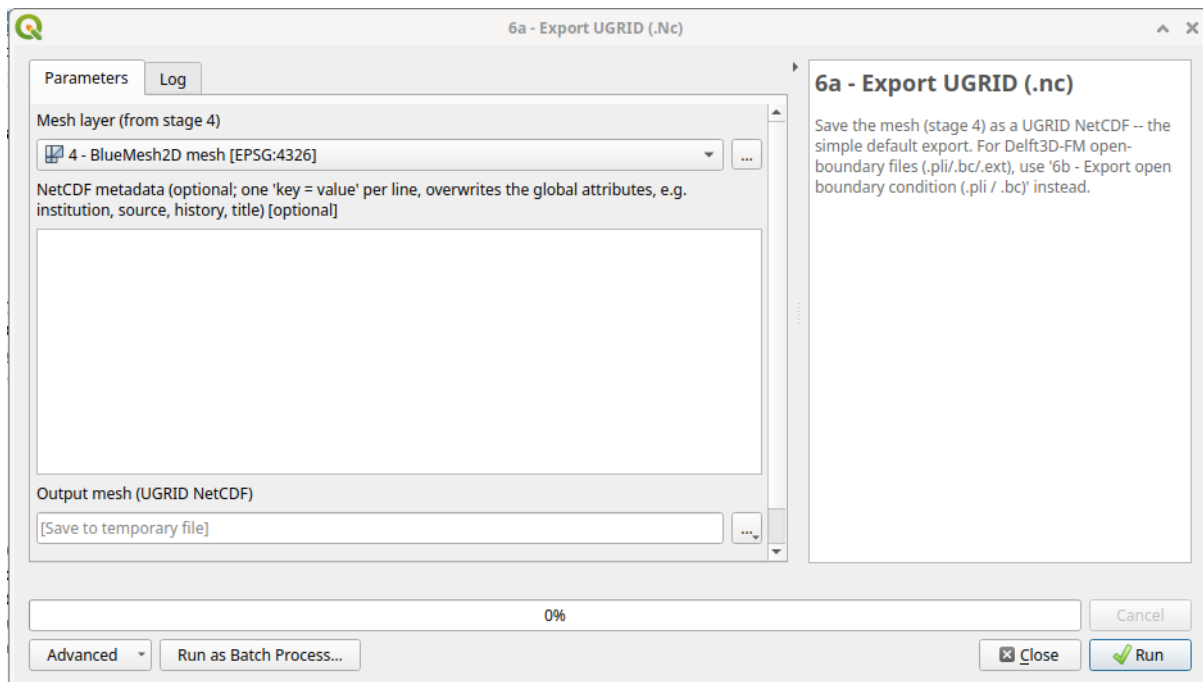


Boundary types can be edited individually by modifying the btype attribute. Multiple boundary nodes can also be edited simultaneously by enabling Show Selected Features, selecting the desired nodes on the map, and using Update Filtered in the attribute table. Set the btype value to 'closed' for closed boundaries or 'open' for open boundaries.

6 - Export

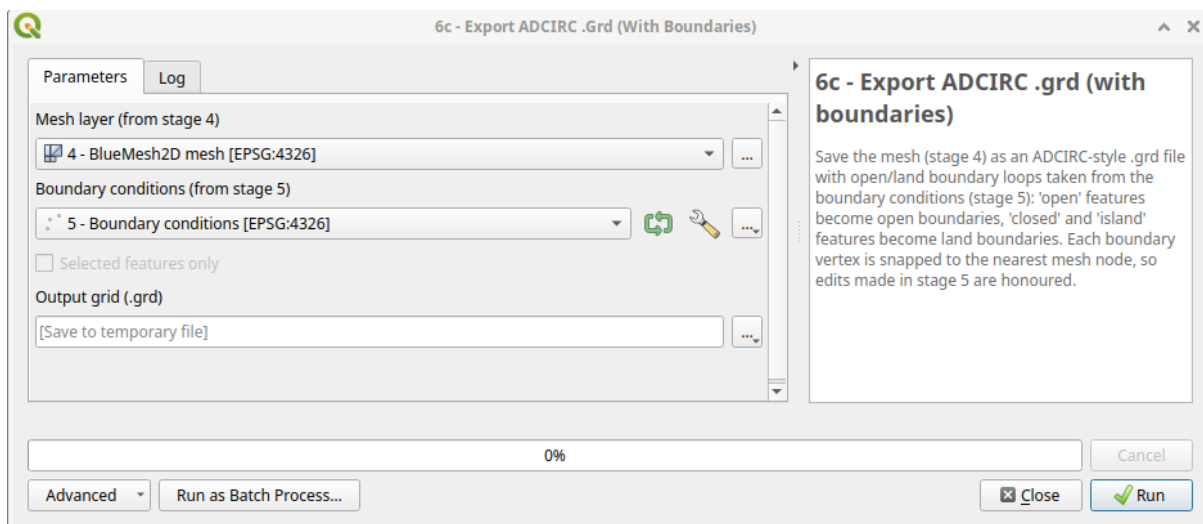


The final mesh is shown above.



The mesh can be exported as a UGRID-compliant NetCDF (.nc) file in the Delft3D-FM format. The metadata can be customized in the dedicated panel before exporting. Open boundary conditions can also be exported using 6b – Export Open Boundary Conditions.

<https://ugrid-conventions.github.io/ugrid-conventions/>



The final mesh can also be exported as an ADCIRC .grd file, including the corresponding boundary conditions.

<https://adcirc.org/home/documentation/users-manual-v53/input-file-descriptions/adcirc-grid-and-boundary-information-file-for-14/>